

Experiment No. 1 (A)

Aim: - Basic safety precaution, Introduction and use of measuring instruments – voltmeter, ammeter, multi-meter.

APPARATUS REQUIRED:-

S. No.	Name of apparatus	Type	Range	Quality
1	Ammeter	MI, MC	(0-2)A AC, (0-1) A DC	As required
2	Voltmeter	MI, MC	(0-300)V AC, (0-20) V DC	As required
3	Multi meter	DIGITAL		1
4	Resistors	Fixed, Variable		As required
5	Capacitors	Fixed, Variable		As required
6	Inductors	Fixed, Variable		As required

THEORY:-

Ammeter:-

Ammeter is an electronic instruments device used to determine the electric current flowing through a circuit. Ammeters measuring current in Milli-ampere range is known as Milli-ammeters. Common types of Ammeters are moving-coil ammeter and moving-iron ammeter. Ammeters are connected in series to the circuit whose current is to be measured. Hence this electronic instruments are designed to have as minimum resistance/loading as possible.

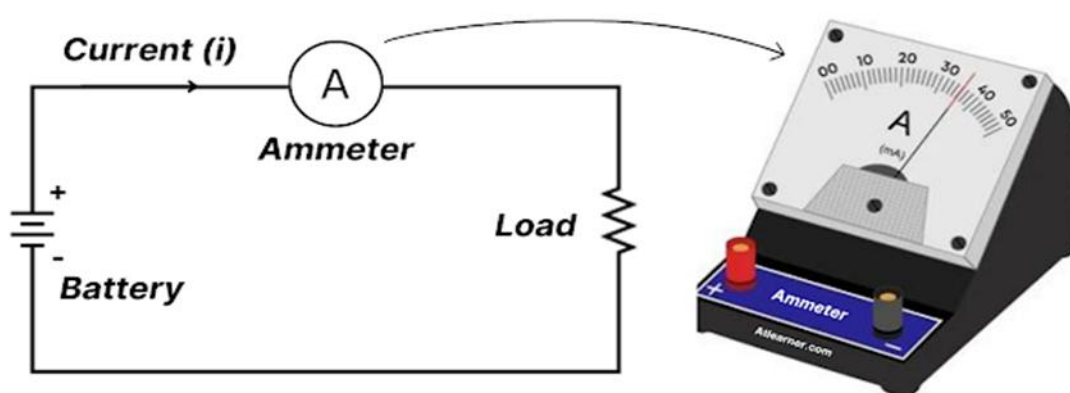


Fig. 1(a).1 Ammeter- Electronic Instruments

Moving Coil Ammeter:-

Moving coil ammeters are used to measure DC Currents. This electronic instruments consists of a coil suspended by two hair springs. This coil is placed in a magnetic field created by a fixed permanent magnet. A torque is experienced when current passes through this coil which is proportional to the current. When the coil turns, the springs will exert a restoring force proportional to the angle turned. By these two forces, the coil will stop at some point and the angular deflection will be proportional to the current.

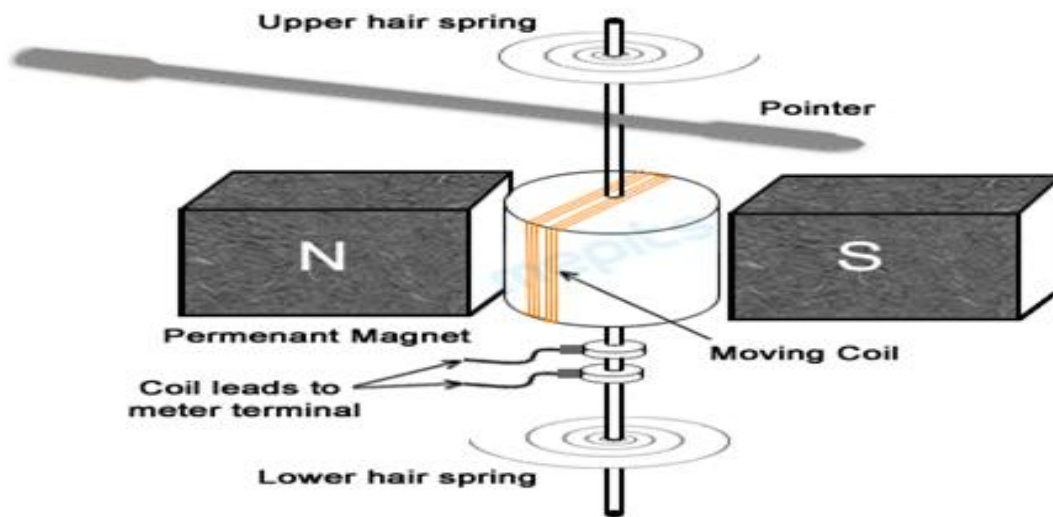


Fig. 1(a).2 Moving coil ammeter

Moving Iron Ammeter:-

Moving iron ammeters as electronic instruments can be used for measuring both direct and alternating currents in electronics lab. In this type of ammeter, a piece of soft iron is used. This iron piece constitutes of a moving vane and a fixed vane. Current to be checked flows through a fixed coil placed around the iron piece. This coil produces a magnetic field proportional to the current. So the iron pieces will get magnetized with the same polarity. The movable vane turns away from the fixed vane due to magnetic repulsion. As the iron turns, the spring of the electronic instruments will exert a restoring force and stop the vane, when both the forces become equal. The pointer of the ammeter is attached to the movable vane, which will point to the proper current reading using a calibrated scale.

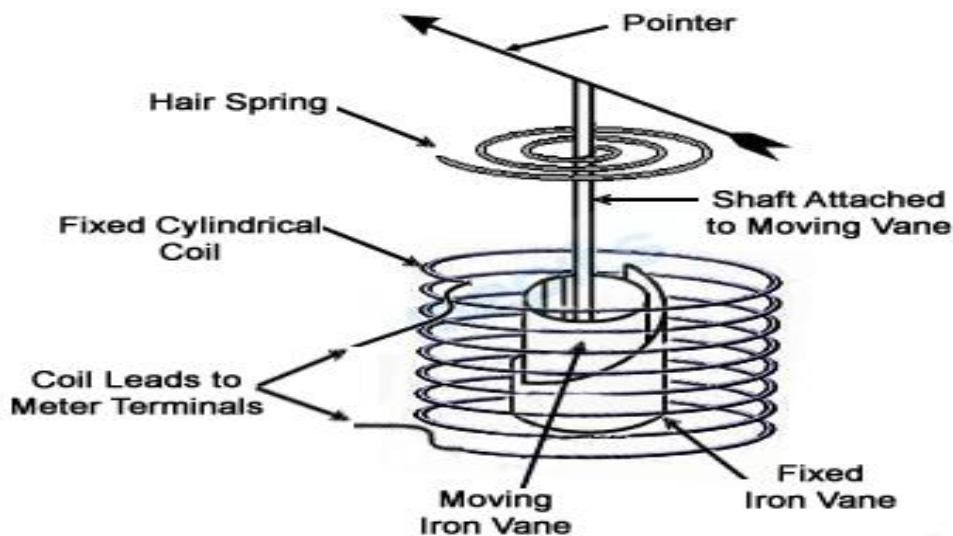


Fig. 1(a).3 Moving Iron Ammeter

Voltmeter:-

Voltmeter is an electronic instruments used in an electric circuit to determine the potential difference or voltage between two different points. Digital and analog voltmeters are available in electronics lab. They are usually connected in parallel (shunt) to the circuit. Hence they are designed to have maximum resistance as possible to reduce the loading effect. This device is also common in electronics lab.

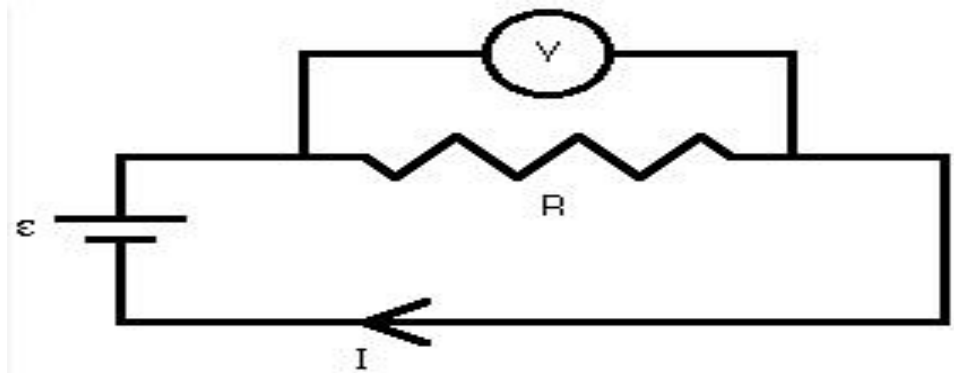


Fig.1 (a).4 (i) Analog Voltmeter



(ii) Digital Voltmeter

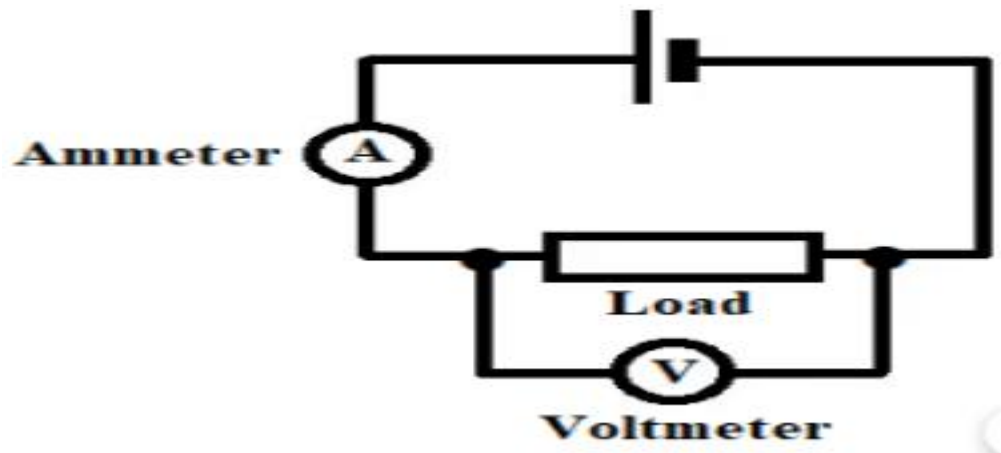
Analog Voltmeter:-

Analog voltmeter is a type of voltmeter and electronic instruments with an extra connection of a series resistor (high resistance). It consists of a movable coil placed in a magnetic field. The coil ends are connected to the measuring terminals. As current flows across the coil, it will start turning due to magnetic force exerted on the coil and thus the hair spring will stop the coil by an equal and opposite restoring force. Angular rotation will be proportional to the voltage in this electronic instruments.

Digital Voltmeter:-

Digital voltmeters can measure both AC and DC measurements with high accuracy as an electronics instrument. It can measure a high voltage up to 1 kV. Main component of a digital voltmeter is an Analog to Digital Converter (ADC). Voltage to be measured is amplified or attenuated properly by the circuit and the output is sent to an Analog to Digital Converter (ADC) IC. This IC will convert the analog signal input to digital signal output. A digital display driven by this IC will display the proper voltage value.

Ammeter with Voltmeter Connection Diagram



Digital Multi Meter (DMM):-

A multi meter or a multi tester, also known as a VOM (Volt-Ohm meter), is an electronic measuring instrument that combines several measurement functions in one unit. A typical multi meter would include basic features such as the ability to measure voltage, current, and resistance. Analog multi meters use a micro ammeter whose pointer moves over a scale calibrated for all the different measurements that can be made. Digital multi meters (DMM, DVOM) display the measured value in numerals, and may also display a bar of a length proportional to the quantity being measured. Digital multi meters are now far more common than analog ones, but analog multi meters are still preferable in some cases, for example when monitoring a rapidly-varying value.

A multi meter can be a hand-held device useful for basic fault finding and field service work, or a bench instrument which can measure to a very high degree of accuracy. They can be used to troubleshoot electrical problems in a wide array of industrial and household devices such as electronic equipment, motor controls, Domestic appliances, power supplies, and wiring systems.

Operation:-

A multi meter is a combination of a multi range DC voltmeter, multi range AC voltmeter, multi range ammeter, and multi range ohmmeter. An un-amplified analog multi meter combines a meter movement, range resistors and switches.

For an analog meter movement, DC voltage is measured with a series resistor connected between the meter movement and the circuit under test. A set of switches allows greater resistance to be inserted for higher voltage ranges. The product of the basic full-scale deflection current of the movement, and the sum of the series resistance and the movement's own resistance, gives the full-scale voltage of the range. As an example, a meter movement that required 1 mill ampere for full scale deflection, with an internal resistance of 500 ohms, would, on a 10-volt range of the multi meter, have 9,500 ohms of series resistance. For analog current ranges, low-resistance shunts are connected in parallel with the meter movement to divert most of the current around the coil.

Again for the case of a hypothetical 1 mA, 500 ohm movement on a 1 Ampere range, the shunt resistance would be just over 0.5 ohms.

Moving coil instruments respond only to the average value of the current through them. To measure alternating current, a rectifier diode is inserted in the circuit so that the average value of current is non-zero. Since the rectified average value and the root-mean-square value of a

waveform need not be the same, simple rectifier-type circuits may only be accurate for sinusoidal waveforms. Other wave shapes require a different Calibration factor to relate RMS and average value. Since practical rectifiers have non-zero voltage drop, accuracy and sensitivity is poor at low values.

To measure resistance, a small battery within the instrument passes a current through the device under test and the meter coil. Since the current available depends on the state of charge of the battery, a multi meter usually has an adjustment for the ohms scale to zero it. In the usual circuit found in analog multi meters, the meter deflection is inversely proportional to the resistance; so full-scale is 0 ohms, and high resistance corresponds to smaller deflections. The ohms scale is compressed, so resolution is better at lower resistance values.

Amplified instruments simplify the design of the series and shunt resistor networks. The internal resistance of the coil is decoupled from the selection of the series and shunt range resistors; the series network becomes a voltage divider. Where AC measurements are required, the rectifier can be placed after the amplifier stage, improving precision at low range.



Fig 1(a).5 Digital Multi meter

Digital instruments, which necessarily incorporate amplifiers, use the same principles as analog Instruments for range resistors. For resistance measurements, usually a small constant current is passed through the device under test and the digital multi meter reads the resultant voltage drop; this eliminates the scale compression found in analog meters, but requires a source of significant current. An auto ranging digital multi meter can automatically adjust the scaling network so that

the measurement uses the full precision of the A/D converter. In all types of multi meters, the quality of the switching elements is critical to stable and accurate measurements. Stability of the resistors is a limiting factor in the long-term accuracy and precision of the instrument.

Quantities measured:-

Contemporary multi meters can measure many quantities. The common ones are:-

- Voltage, alternating and direct, in volts.
- Current alternating and direct in amperes.

The frequency range for which AC measurements are accurate must be specified.

- Resistance in ohms.

Additionally, some multi meters measure:-

- Capacitance in farads.
- Conductance in Siemens.
- Decibels.
- Duty cycle as a percentage.
- Frequency in hertz.
- Inductance in henrys.
- Temperature in degrees Celsius or Fahrenheit, with an appropriate temperature test probe, often a thermocouple.

Digital multi meters may also include circuits for:

- Continuity tester; sounds when a circuit conducts
- Diodes (measuring forward drop of diode junctions), and transistors (measuring current gain and other parameters)
- Battery checking for simple 1.5 volt and 9 volt batteries. This is a current loaded voltage scale which simulates in-use voltage measurement.

PRECAUTIONS:-

1. All connection must be tight.
2. Get the circuit connections checked by the teacher before performing the experiment.
3. Power to the circuit must be switched on in the presence of the teacher.
4. Get the experimental readings checked by the teacher.
5. Don't touch directly the live parts of equipment and circuit.
6. Wear leather shoes in the lab.

Result: - We have studied about the construction, working and connection of moving coil, moving iron, voltmeter, ammeter and multi meter.

VIVA VOICE:-

- 1) What are the various uses of multi meter?
- 2) What is a Ammeter?
- 3) Which device is use to measure ac current?
- 4) What is a voltmeter?